Analytical Chemistry Cumulative Examination
April 17, 1999

Please answer the following questions involving optical spectroscopy. All work must be shown. Be sure to indicate the logic used to solve each problem.

1. A 0.9934-g sample of steel is dissolved in acid and the solution diluted to 250.0 mL (solution B). A 50.00 mL aliquot of solution B is treated with potassium persulfate in the presence of Ag^+ ions acting as a catalyst, and potassium periodate, whereupon Mn and Cr are oxidized to MnO_4^- and $Cr_2O_7^{2-}$ and diluted to 100.0 mL (solution D). The absorbance of solution D at 440 nm is 0.204 and at 545 nm is 0.170 in a 1.00 cm cell. Calculate the percent Mn and Cr in the steel taking into account the following data from the literature. (35 points)

λ, nm	ε MnO_4^-	ε $Cr_2O_7^{2-}$
440	95	369
545	2350	11.0

2. Given that P is defined as the power of the transmitted radiation and P_o is defined as the power of the incident radiation in a measurement obeying Beer's law, complete the following table. (10 points)

P/P_o	%T	A
1.00		
	10.0	
		2.00

3. A 0.0782 g sample of butylamine (m.w. 73.14) is dissolved in water and diluted to 250.0 mL (solution B). A 10.0 mL aliquot of solution B is allowed to react with picric acid and diluted to 100.0 mL (solution D). A 10.00 mL aliquot of solution D is diluted to 100.0 mL and the absorbance of the final solution at 380 nm in a 1.00 cm cell is 0.515. Calculate the percent purity of the butylamine, given that the molar absorptivity ε_{380} of the absorbing compound (butylamine picrate) is 1.35×10^4 L $mol^{-1}cm^{-1}$. (35 points)

4. When setting100% T with a water blank, by mistake the transmittance scale reading was set at 90 instead of 100. After this setting, the transmittance of an unknown solution was 30.8 %. Calculate the real tranmittance of the unknown solution. (10 points)

5. Which of the following relations are false? a) $A=abc_{mol/L}$; b) $T=10^{-A}$; c) $A= \varepsilon bc_{g/L}$; d) $A= \log(P/P_o)$; e) $a= \varepsilon \times MW$ (10 points)

Definitions:

A= absorbance
P_o= power of incident radiation
P = power of transmitted radiation
a = proportionality constant when c is expressed in g/L, absorptivity
b = path length through solution
T = transmittance
ε = proportionality constant when c is expressed in mol/L, molar absorptivity
MW = molecular weight in grams/mol